

# Integrated Circuit ECE326

## Lec05

### Synthesis Digital Circuit

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- ☐ **Transmission Gate**
- ☐ **Synthesize of Multiplexer**
- ☐ **Assignment (01)**



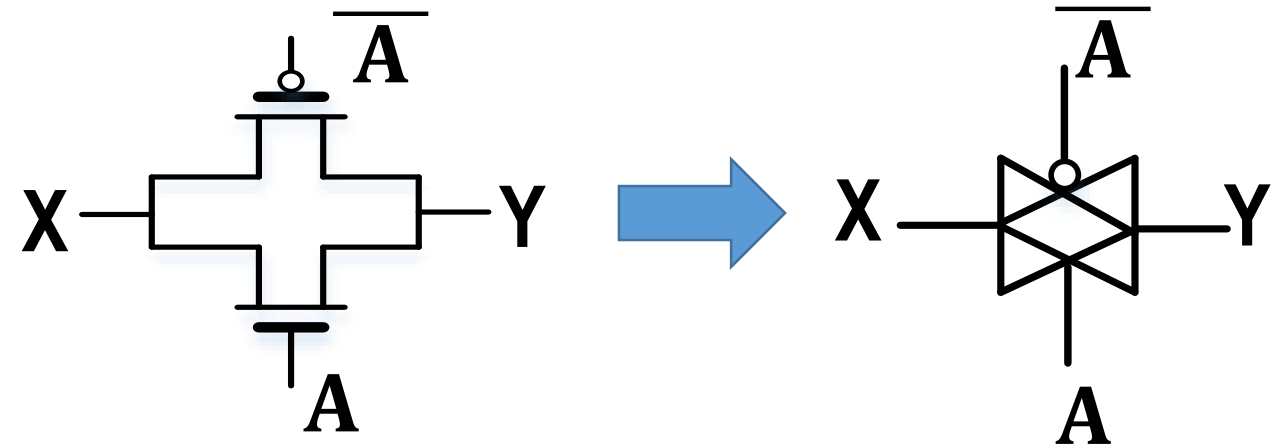
- ☐ **Transmission Gate**
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# Transmission Gate (1/2)



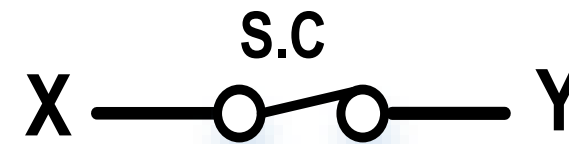
- Transmission Gate (TG) consists of n-channel and p-channel E-MOSFETs which are connected parallel.

|          | Good      | Weak  |
|----------|-----------|-------|
| N-MOSFET | "0"       | "1"   |
| P-MOSFET | "1"       | "0"   |
| TG       | "1" & "0" | ----- |

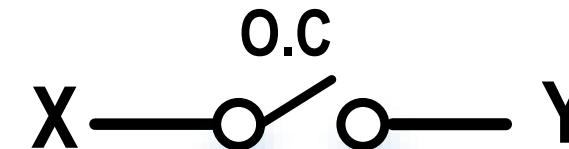


## TG's operation

- @  $A = "1"$  → TG will be Short Circuit (S.C)



- @  $A = "0"$  → TG will be Open Circuit (O.C)



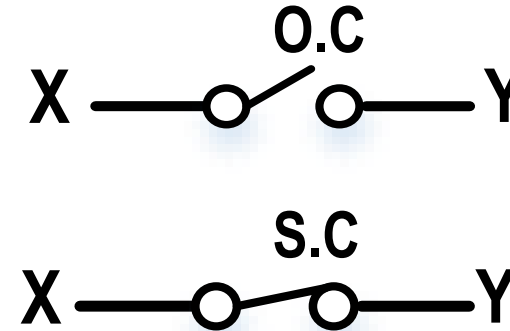
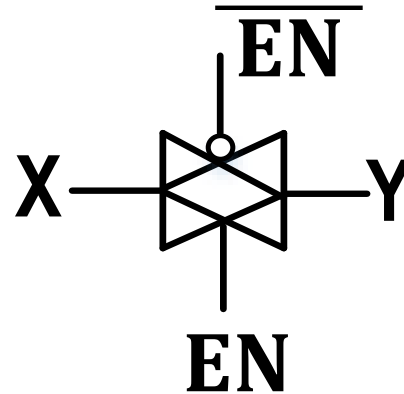
# Transmission Gate (2/2)



## □ Applications of TG

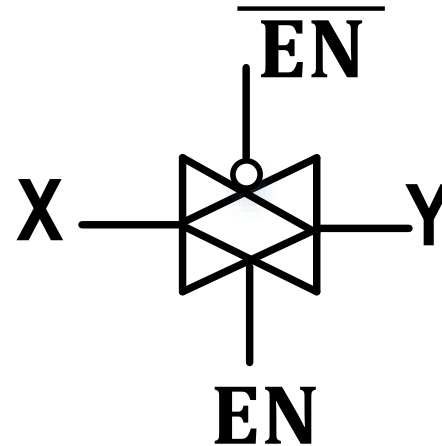
- TG is used to build a multiplexer for transferring data

| EN | Y                     |
|----|-----------------------|
| 0  | Z<br>"High impedance" |
| 1  | Y=X                   |



- TG is used to build a tri-state buffer

| EN | X | Y |
|----|---|---|
| 0  | 0 | Z |
| 0  | 1 | Z |
| 1  | 0 | 0 |
| 1  | 1 | 1 |



## Advantages ☺

- TG is Good in transferring data
- TG is Good in transferring "1" and "0"
- Use only two transistors to build tri-state buffer

## Disadvantages ☹

- Noise on X is passed to Y



- ❑ **Transmission Gate**
- ❑ **Synthesize of Multiplexer**
- ❑ **Assignment (01)**

# Synthesize of Multiplexer (1/9)



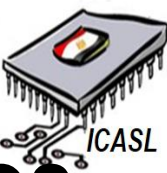
Ex.1. Synthesize a 4\*1 multiplexer using a) Transmission gates b) CMOS transistors c) Universal gates d) Which design is more efficient? why?

**Ans.**

a) Using transmission gates

- ☐ Step1: Determine # of inputs, # of outputs, and # of selectors
- ☐ Step2: Build the truth table to get the Mux's function
- ☐ Step3: Build a grid
  - ☐ Draw horizontal lines → # of inputs
  - ☐ Draw vertical lines → # of selectors
- ☐ Step4: Placement the transmission gates in the cross points.
- ☐ Step5: Routing the transmission gates with the other wires.

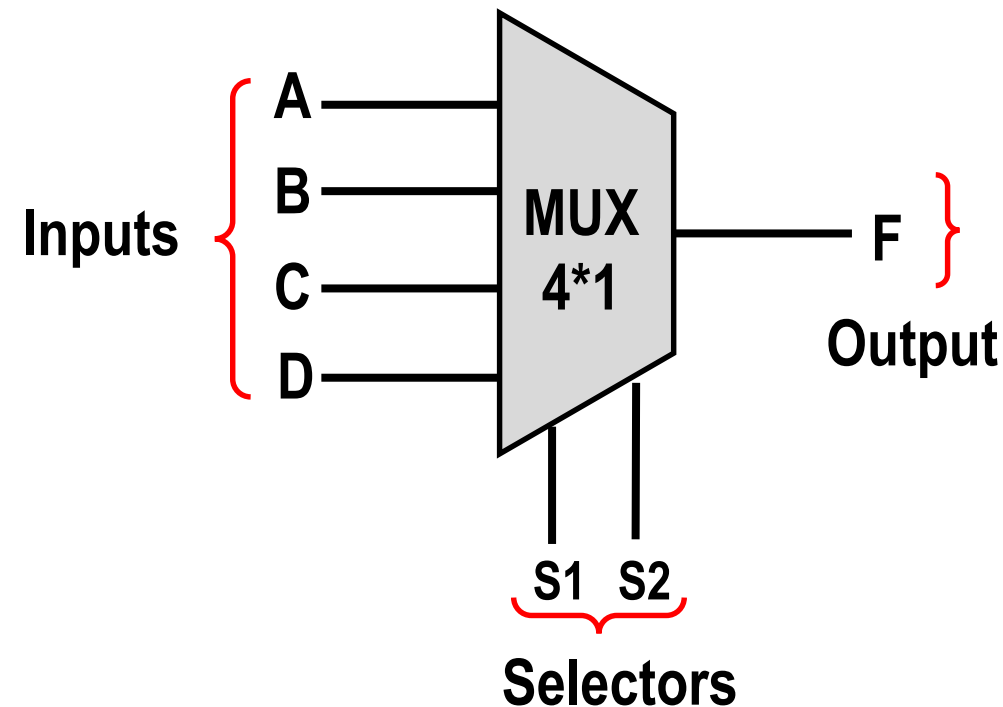
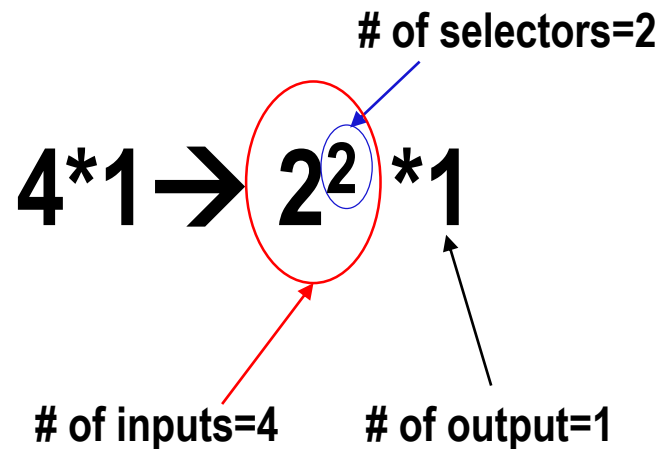
# Synthesize of Multiplexer (2/9)



Ex.1. Synthesize a 4\*1 multiplexer using a) Transmission gates b) CMOS transistors c) Universal gates d) Which design is more efficient? why?

**Ans.**  
a) Using transmission gates

□ Step1: Determine # of inputs, # of outputs, and # of selectors





# Synthesize of Multiplexer (3/9)



Ex.1. Synthesize a 4\*1 multiplexer using a) Transmission gates b) CMOS transistors c) Universal gates d) Which design is more efficient? why?

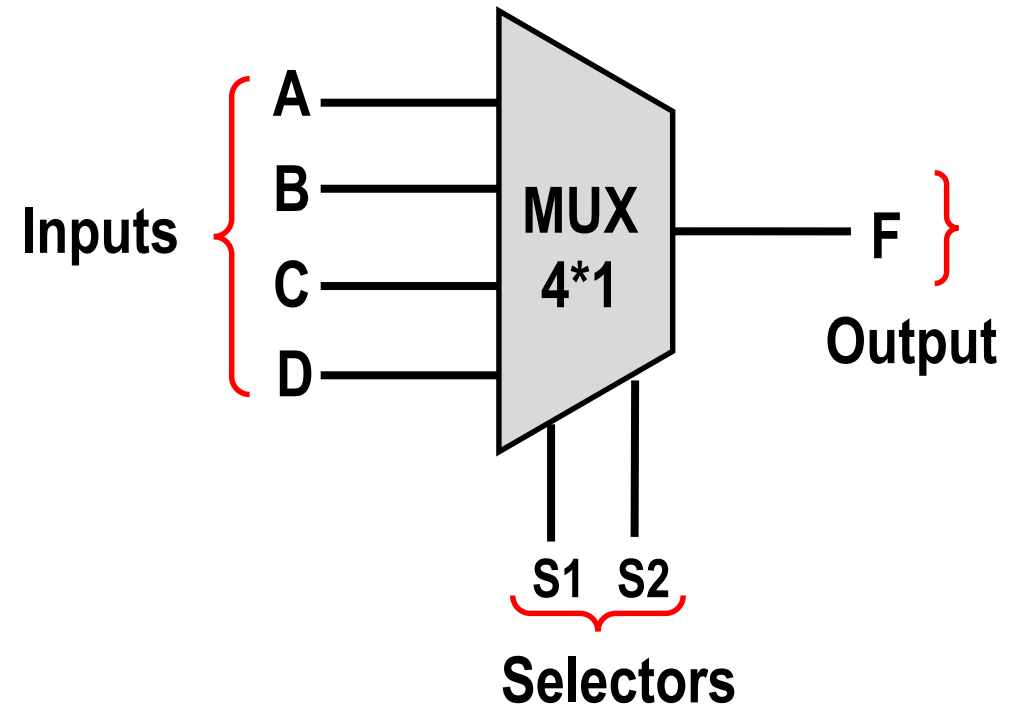
**Ans.**  
a) Using transmission gates

□ Step2: Build the truth table to get the Mux's function

Truth table

| Selectors |    | Output |
|-----------|----|--------|
| S2        | S1 | F      |
| 0         | 0  | A      |
| 0         | 1  | B      |
| 1         | 0  | C      |
| 1         | 1  | D      |

$$F = A \cdot \overline{S_1} \cdot \overline{S_2} + B \cdot \overline{S_1} \cdot S_2 + C \cdot S_1 \cdot \overline{S_2} + D \cdot S_1 \cdot S_2$$



# Synthesize of Multiplexer (4/9)



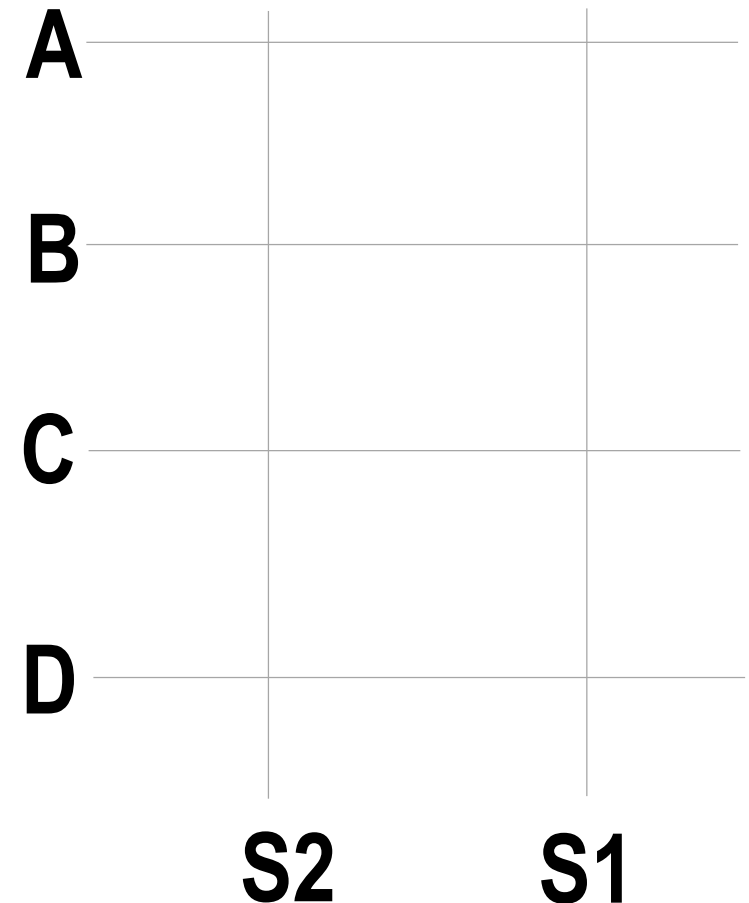
Ex.1. Synthesize a 4\*1 multiplexer using a) Transmission gates b) CMOS transistors c) Universal gates d) Which design is more efficient? why?

**Ans.**  
a) Using transmission gates

☐ Step3: Build a **grid**

☐ Draw horizontal lines → # of inputs

☐ Draw vertical lines → # of selectors



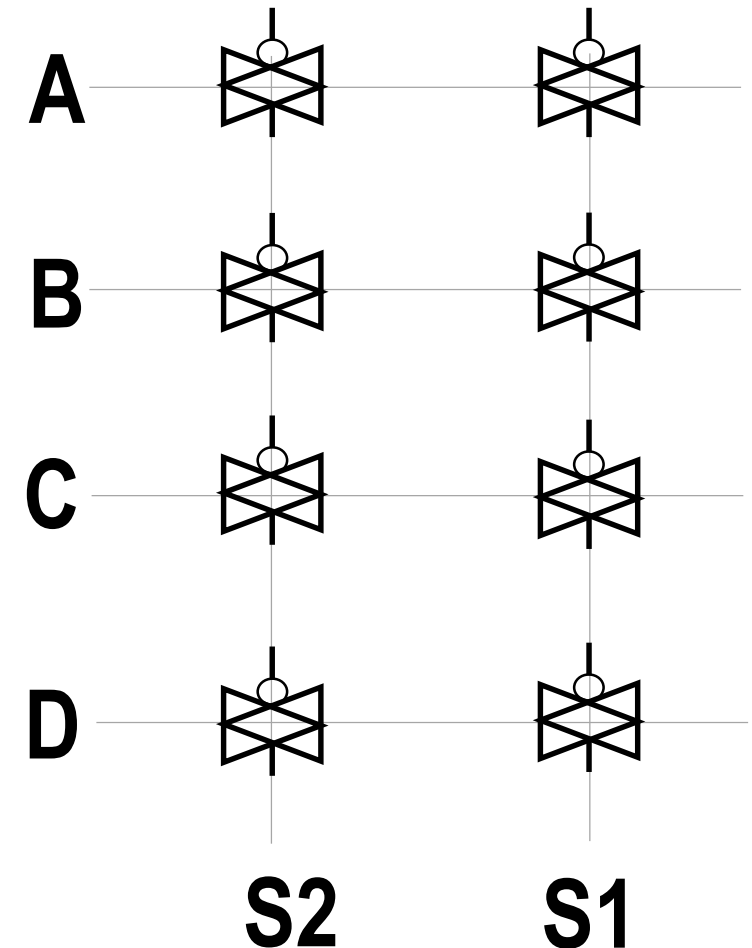
# Synthesize of Multiplexer (5/9)



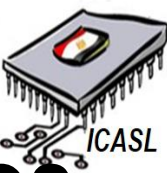
Ex.1. Synthesize a 4\*1 multiplexer using a) Transmission gates b) CMOS transistors c) Universal gates d) Which design is more efficient? why?

**Ans.**  
a) Using transmission gates

□ Step4: **Placement** the transmission gates in the cross points.



# Synthesize of Multiplexer (6/9)



Ex.1. Synthesize a 4\*1 multiplexer using a) Transmission gates b) CMOS transistors c) Universal gates d) Which design is more efficient? why?

**Ans.**  
a) Using transmission gates

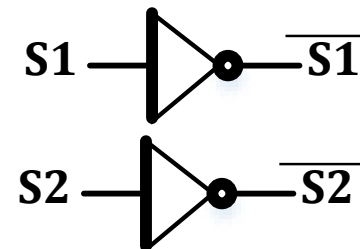
❑ Step5: Routing the transmission gates with the other wires.

$$F = A \cdot \overline{S_1} \cdot \overline{S_2} + B \cdot S_1 \cdot \overline{S_2} + C \cdot \overline{S_1} \cdot S_2 + D \cdot S_1 \cdot S_2$$

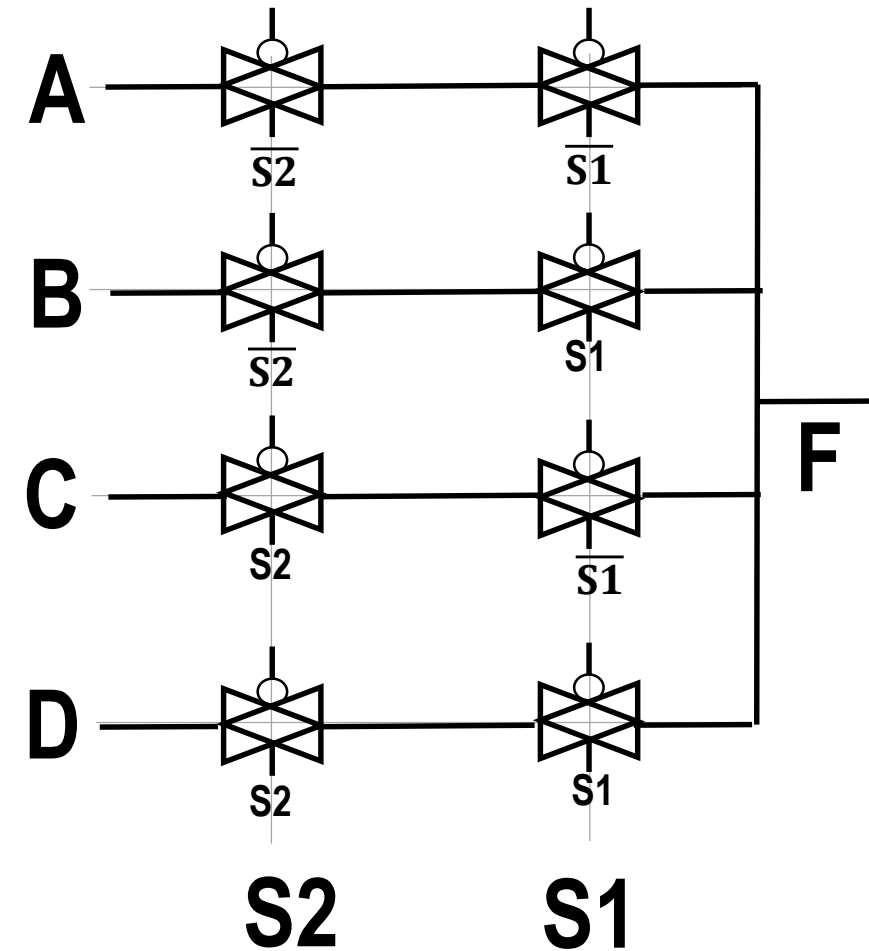
# of transistors =  $8 \cdot 2 + 2 \cdot 2 = 20$  😊

# of TGs

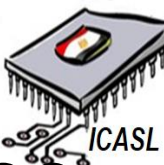
# of inverters



Do not Forget the Inverters



# Synthesize of Multiplexer (7/9)



Ex.1. Synthesize a 4\*1 multiplexer using a) Transmission gates b) CMOS transistors c) Universal gates d) Which design is more efficient? why?

**Ans.**  
**b) Using CMOS transistors**

# of variables  $> 4$ ,  
you do not need K-Map

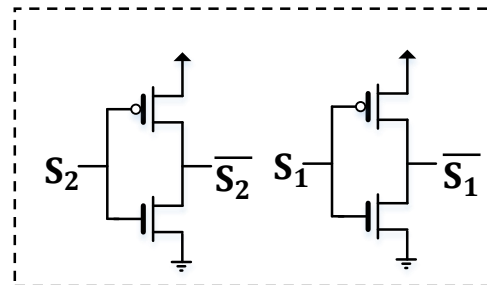
$$F = A \cdot \overline{S_1} \cdot \overline{S_2} + B \cdot S_1 \cdot \overline{S_2} + C \cdot \overline{S_1} \cdot S_2 + D \cdot S_1 \cdot S_2$$

$$\overline{F} = \overline{A \cdot \overline{S_1} \cdot \overline{S_2} + B \cdot S_1 \cdot \overline{S_2} + C \cdot \overline{S_1} \cdot S_2 + D \cdot S_1 \cdot S_2}$$

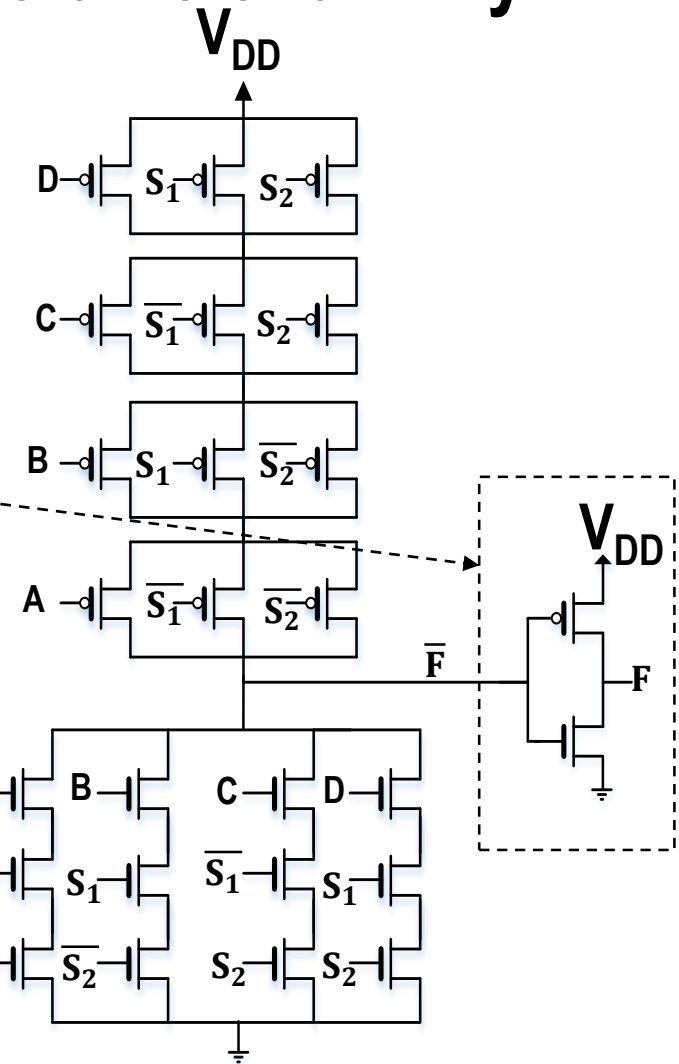
Series      Series      Series      Series

Parallel

$$\# \text{ of transistors} = 12 \cdot 2 + 3 \cdot 2 = 30 \quad \text{😊}$$



Do not Forget the Inverters



# Synthesize of Multiplexer (8/9)



Ex.1. Synthesize a 4\*1 multiplexer using a) Transmission gates b) CMOS transistors c) Universal gates d) Which design is more efficient? why?

**Ans.**  
c) Using Universal gates

$$F = A \cdot \overline{S_1} \cdot \overline{S_2} + B \cdot S_1 \cdot \overline{S_2} + C \cdot \overline{S_1} \cdot S_2 + D \cdot S_1 \cdot S_2$$

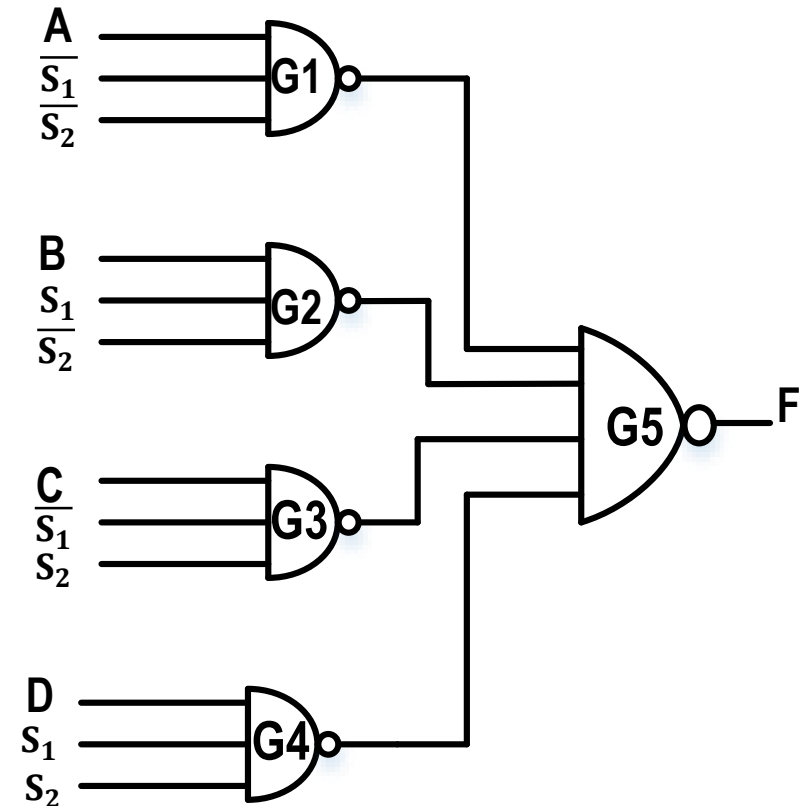
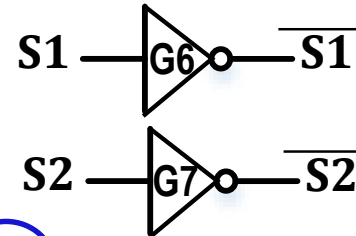
Due to the function is in form of **SOP**, NAND gates are used.

$$\overline{\overline{F}} = \overline{A \cdot \overline{S_1} \cdot \overline{S_2} + B \cdot S_1 \cdot \overline{S_2} + C \cdot \overline{S_1} \cdot S_2 + D \cdot S_1 \cdot S_2}$$

$$\overline{\overline{F}} = \underbrace{A \cdot \overline{S_1} \cdot \overline{S_2}}_{G_1} \cdot \underbrace{B \cdot S_1 \cdot \overline{S_2}}_{G_2} \cdot \underbrace{C \cdot \overline{S_1} \cdot S_2}_{G_3} \cdot \underbrace{D \cdot S_1 \cdot S_2}_{G_4}$$

$G_5$

# of transistors =  $4 \cdot 3 \cdot 2 + 1 \cdot 4 \cdot 2 + 2 \cdot 2 = 36$  😊



# Synthesize of Multiplexer (9/9)



Ex.1. Synthesize a 4\*1 multiplexer using a) Transmission gates b) CMOS transistors c) Universal gates d) Which design is more efficient? why?

**Ans.**  
d) Which design is more efficient? why?

|                  | Transmission gates | CMOS transistors | Universal gates |
|------------------|--------------------|------------------|-----------------|
| # of transistors | 20 😊               | 30               | 36              |

Synthesizing by Transmission gates for MUX requires low number of transistors,  
Thus, the consumption power is low and the consumed silicon area is small



- ☐ **Transmission Gate**
- ☐ **Synthesize of Multiplexer**
- ☐ **Assignment (01)**



# Assignment (01)



**A.1. Synthesis a 2\*1 multiplexer using min. number of transistor.**

- Clarify your design using cadence.
- The 1<sup>st</sup> channel should be ECG or PPG signal(hint. You will use PWL source)
- The 2<sup>nd</sup> channel should be sin wave with amplitude of 0.5V and frequency of 100Hz signal (hint. You will use sin source)

**Deadline: 15/3/ 2024**

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# Thanks!